



Portable gamma spectrometry for rapid assessment of soil texture, organic carbon and total nitrogen in agricultural soils

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Abstract

Purpose In the context of developing interest in regenerative farming to enhance soil health and wider ecosystem service benefits, this study aimed to evaluate the potential for using a portable gamma sensor (Medusa MS-700) to survey soil health parameters in agricultural fields.

Materials and methods A Medusa MS-700 portable gamma spectrometer was deployed to evaluate activity concentrations of ⁴⁰K, ²³⁸U, ²³²Th and ¹³⁷Cs across agricultural fields. In parallel, 90 soil samples were collected for laboratory analysis of SOM by loss on ignition (LOI), total C (TC) and N (TN), particle size, moisture content and bulk density. Linear regression was used to derive prediction models of soil parameters based upon radionuclide concentrations.

Results and discussion Correlation between soil parameters and radionuclide concentrations showed negative associations between radionuclides and moisture content, LOI, TC and TN, and most notably for ⁴⁰K ($r_s > -0.8$). This is likely related to (i) the diluting effect of organic matter in relation to the gamma-emitting parent material and (ii) the attenuating effect of soil moisture upon gamma rays. Correlations with texture properties were generally less strong with coefficients < 0.7 . Linear regression prediction models returned mean absolute errors (MAE) of around 1% TC and 0.1% TN.

Conclusion Findings show the potential for using this portable gamma sensor for rapid spatial analysis of TC and TN across agricultural land units. Considering the gamma sensor as a decision support tool, we discuss the potential for rapid detection across wide spatial scales to inform targeted measures for regenerative action. From a land management perspective, prediction models from this dataset are encouraging wherein site-specific validation of sensor data is essential.

Keywords Portable gamma spectrometry · Soil health · Soil organic carbon · Sustainable land management

1 Introduction

It is widely recognised that global soil resources are under threat from management practices, which destabilise soil systems, further exacerbated by climate change and related

extreme weather patterns (Nearing et al. 2004; FAO & ITPS 2015; IPCC 2021). Resulting soil degradation undermines soil functions and threatens the wide range of ecosystems services provided by healthy soil systems (Kihara et al. 2020). Indeed, there is an increasing appreciation of the role of healthy soil systems in food, water and energy security, and attention to soil management *is essential* for achieving many of the United Nations Sustainable Development Goals (Lal et al. 2021).

Numerous studies highlight soil organic carbon (SOC) as a pivotal factor in soil health since it underpins a wide range of soil functions related to nutrient and water cycling, soil structure and biodiversity, with its loss often central to degradation processes (FAO 2017; Laban et al. 2019; Taylor et al. 2021). Attention to maintaining and restoring soil resources and associated SOC could, therefore, support

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community resilience to extreme weather events and even mitigate climate change if carbon sequestration potential is recognised (IPCC 2019; Stewart et al. 2020). SOC pools are highlighted as crucial for climate change adaptation strategies across the globe, not only in arid and semi-arid environments (FAO 2017; Liniger et al. 2011; Middelndorf et al. 2017), but also in temperate regions where agricultural intensification has exerted pressure on soil resources. In the UK, for example, greater emphasis on soil health assessment has been introduced as part of a new Environmental Land Management Scheme (ELMS), which includes assessment of SOC (as a component of organic matter) and soil structure (Defra 2020).

It follows, then, that monitoring soil health parameters, such as SOC, is valuable for underpinning sustainable land management practices both in terms of identifying areas of degradation to enable targeted mitigation, and to determine the relative effects of differing land use approaches. Increasingly, the importance of recognising the connectivity of land units in *whole-systems* approaches to soil conservation has underpinned the use of remote and proximal sensing tools to achieve improved spatial resolution relative to traditional soil sampling techniques (Blake et al. 2020; Andries et al. 2021). These approaches rely on the measurement of proxies, which have been characterised against a particular soil parameter of interest. Of these, the activity concentration (Bq/kg) of naturally occurring gamma-ray emitting radionuclides has been widely used in soil and sediment studies to determine soil redistribution rates and soil health parameters (Dercon et al. 2012; Reinhardt and Herrmann 2019). Advances in portable gamma spectrometry offer an opportunity for high spatial resolution surveys of radionuclide proxies in situ, which are potentially advantageous over time-consuming and costly laboratory-based analyses. Portable gamma spectrometers are capable of very rapid data acquisition (1 Hz) when compared to other in situ techniques, such as near infra-red spectroscopy (NIRS) and, therefore, provide an opportunity for continuous monitoring (by foot or vehicle) across broad spatial scales (van der Veeke et al. 2021). Furthermore, as a proximal technique spatial resolution can be easily controlled by the user, with surveys tailored to land management scenarios at the plot, farm or landscape scales.

In soils, the naturally occurring radionuclides ^{40}K ($t_{1/2} = 1.3 \times 10^9 \text{y}$), ^{238}U ($t_{1/2} = 4.46 \times 10^9 \text{y}$) and ^{232}Th ($t_{1/2} = 1.39 \times 10^9 \text{y}$) are present in sufficient concentrations to enable their activity concentrations to be detected and quantified through the use of portable gamma spectrometry (Reinhardt and Herrmann 2019). In addition to these geogenic sources, background radioactivity in soils is further enhanced via fallout radionuclides (FRN) such as ^{137}Cs ($t_{1/2} = 30 \text{y}$), which is an artificial radioisotope widely distributed across the globe as a result of atmospheric release from weapons testing and the 1986 Chernobyl disaster.

The shape of the spectrum resulting from the decay of ^{40}K , ^{238}U , ^{232}Th and ^{137}Cs is sufficiently unique to enable their respective origins to be derived. Portable gamma detectors are commonly deployed as vehicle-mounted units for ground-based surveys at the field-scale or mounted to unmanned aerial vehicles (UAV) for landscape-scale surveys (Reinhardt and Herrmann 2019). Measurement radius and measured soil volume increase with increasing detector height from the ground, with most of the gamma intensity originating from around 0.3 m depth at a detector height of 0.8 m with respect to ^{232}Th . When considering peak emission at 2620 keV, ^{232}Th measurement can be expected to represent a larger fraction of soil volume than lower peak energy emitters such as ^{40}K (van der Veeke et al. 2021). In all cases, the largest fraction of the detected gamma radiation originates from the soil column directly below the detector. Detected gamma intensity reduces with increasing height of the detector above ground as a function of energy such that intensity decrease follows $^{40}\text{K} > ^{238}\text{U} > ^{232}\text{Th}$ (van der Veeke et al. 2021).

To date, applications of proximal gamma spectrometry to inform land management decisions are not widespread, although available evidence does suggest potential for the tool to be used to determine soil properties to support farm and landscape-scale decision-making. For example, some studies demonstrate the relationship between radionuclide concentrations and soil particle size, highlighting the potential for soil texture mapping (e.g. Heggemann et al. 2017; Pätzold et al. 2020). These relationships are underpinned by the large specific surface area (and thus sorption capacity) and mineralogical structure of clay particles. Other studies demonstrate relationships between radionuclide concentrations and SOC and total nitrogen (Rawlins et al. 2009; Dierke and Werban 2013; Mahmood et al. 2013). Given the geogenic origin of ^{40}K , ^{238}U and ^{232}Th , relationships with SOC and nitrogen are based upon indirect effects of soil organic matter (SOM) upon gamma signals. It is most likely that organic matter has both a diluting and attenuating effect upon gamma emission, with attenuation particularly affected by moisture held within organic material (Franz et al. 2017; Reinhardt and Herrmann 2019). Relationships appear to be highly site-specific, and the development of universal prediction models is likely to be dependent upon the extent of spatial heterogeneity of soil mineralogy (Pätzold et al. 2020). Nevertheless, the development of site-specific prediction models appears plausible and represents an important initial step for exploring radionuclide-soil parameter relationships and developing a dataset for application at broader spatial scales. To the best of the authors' knowledge, to date, research relating to the use of portable gamma spectrometry in a range of land management settings is sparse.

Against this background, we detail the use of portable gamma spectrometry (with a commercially available

Medusa Radiometrics MS-700 gamma-ray spectrometer) for predicting soil texture and organic matter-related parameters (organic carbon and nitrogen) in agricultural soils in South-west UK. We explore the use of the tool in its simplest form, with minimal data processing in a 'ready to use' manner, to provide site-specific data that is accessible to the end user, i.e. the farmer or farm advisor. This contribution presents the findings of important foundation research, which aimed to determine the relationships between radionuclide activity concentrations and soil health parameters at two agricultural field locations, and provides the basis for discussion relating to the use of the approach for decision support across a critical period of agricultural transition.

2 Materials and methods

2.1 Study site

Field surveys were undertaken in the winter–spring period of 2022 at the Duchy College West Coombeshead Farm in east Cornwall, UK (50°32'33.01"N; 4°19'34.98"W). The field sites were at an elevation of ~160 m (amsl) receiving an average annual rainfall of around 1300 mm. Soils in the area are typically free-draining, slightly acid loams overlying Brendon Formation slate, siltstone and sandstone. Improved pasture is typical of this area, with some fields allocated to arable rotations for the production of fodder crops. Two fields in close proximity (< 1 km) were selected for the survey to capture a range of potential SOM values, one of

permanent pasture (5.6 ha area) and another representing a rotation site, which was in a temporary grass ley at the time of the study (10 ha area). Surveys were undertaken during dry weather with no rainfall received at the sites in at least 3 days prior to the survey.

2.2 Soil survey and sampling

For robust determination of site-specific relationships between radionuclide activity concentration and soil parameters, it is crucial that sampling and processing procedures are standardised to reduce uncertainties as far as is practicable.

In situ gamma measurements were carried out using a MS-700 portable gamma-ray spectrometer (Medusa Radiometrics, Netherlands), fitted with a 700 ml CsI scintillator detector. The detector used the mDOS (medusa Detector Operating System) for local data acquisition and handling, with the data acquisition rate set to 1 Hz. Following data collection in the field, full spectrum analysis (Hendriks et al. 2001) was carried out using Medusa Radiometrics algorithms held in Gamman software, which used the detector-specific calibration files to stabilise gamma spectra and provide activity concentration data (Bq/kg). Full details of the calibration procedure are documented by van der Graaf et al. (2011). An example spectrum is shown in Fig. 1. Data processing was carried out using the repeated average approach, which combined and averaged data from every five measurements.

A walking survey was undertaken with the detector held at 0.8 m from the ground. At each field site, gamma

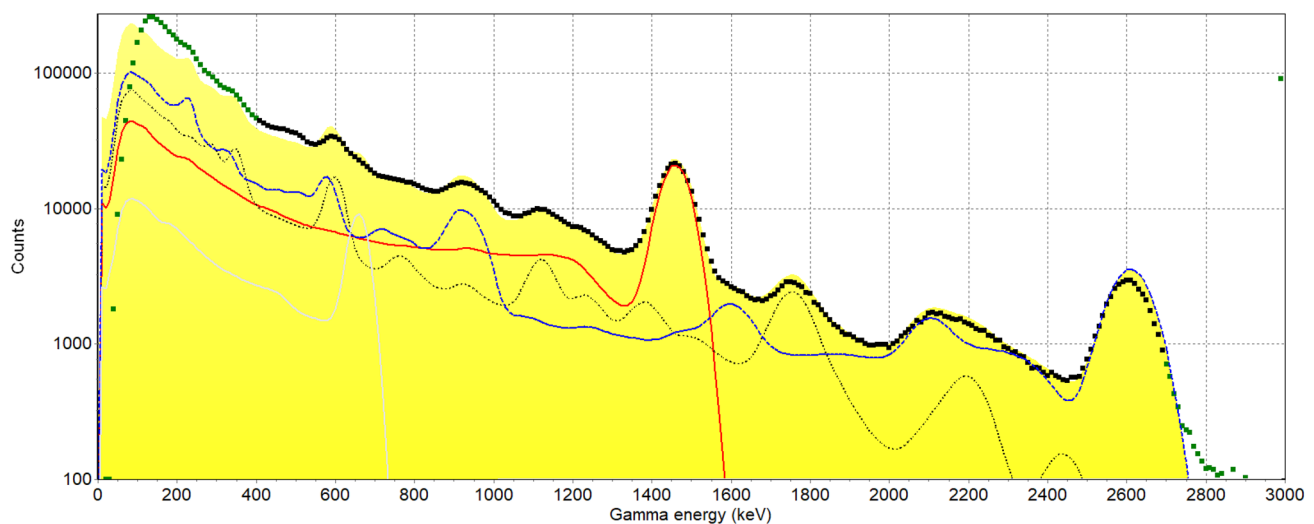


Fig. 1 Example summed spectrum stabilised using Gamman software. Full details of calibration procedures are provided by van der Graaf et al. (2011). The yellow shaded area shows the spectrum peaks fitted to the measured field data (black squares). The contributing radionuclides of interest are ^{40}K (solid red line), ^{232}Th (dashed blue

line) and ^{238}U (black dotted line). The light grey line denotes ^{137}Cs , not utilised in this study owing to low counts. Upper- and lower-level discriminators are indicated by green squares. Counts are in logarithmic scale

measurements were obtained at 20 m intervals along transects, which enabled representative surveys across the field topography. At each survey location, gamma measurements were recorded for 30 s (30 measurements at 1 Hz), before moving to the next location in a ‘stop–go’ survey. In addition, at each point of the gamma measurements, soil cores were taken at depths of up to 0.3 m into the root zone using a stainless-steel Gator Probe (van Walt, UK). Three soil cores were taken from an area of soil below the gamma detector and combined into one spatial composite sample. Separate cores were taken and retained for bulk density measurements (IAEA 2016). In total, 90 sample points were used in the study.

2.3 Laboratory procedures

Fresh (field moist) composite soil samples were lightly disaggregated by hand, mixed and sub samples retained for gravimetric moisture determination. The remaining sample was dried at 40 °C, lightly disaggregated by hand and sieved to < 2 mm (referred to here as the bulk sample). A subsample of the bulk material was retained for particle size analysis. The remaining bulk material was pulverised using a Pulverisette 5 planetary mill (Fritsch, Germany) at 300 rpm for 3 min to ensure < 180 µm particles. Mass loss on ignition (LOI) was undertaken in triplicate at 450 °C for 4 h in a muffle furnace following BS 1377–3:1990. Relative standard deviation (RSD) was shown to be < 2%. Analysis for total C (TC) and total N (TN) was carried out using a vario EL cube (Elementar, Germany). Samples were prepared in triplicate by weighing 10–20 mg of sample into tin capsules. To ensure detection, 15–20 mg of sample was used for those with lower expected C content, and 10–15 mg was used for samples with higher expected C. Approximately 2 mg of acetanilide was used as the calibration standard and analysed three times to establish K-factors. The standard was used during sample runs to check for instrument drift. Analyses of triplicates showed RSD to be < 3% for C and N. Prior analysis of inorganic C (IC) using a Skalar Primacs Carbon Analyser (Skalar Analytical, Netherlands) confirmed that IC was below detection limits, and any C detected was, therefore, deemed to be organic carbon. Gravimetric soil moisture was calculated by drying field moist samples at 105 °C until constant mass. Similarly, bulk density was calculated

using dried (105 °C) core sample mass and core volume (IAEA 2016). The particle size of the mineral soil fraction was measured by laser diffraction following sample oxidation using H₂O₂ to remove organic matter. A Malvern Mastersizer 2000 with Hydro-G (Malvern Instruments Limited, UK) was used (assuming spherical particles) in compliance with ISO 13320:2009.

2.4 Data handling

To assess relationships between radionuclides and soil parameters, data analyses were performed in R Studio (2022.02.0443). An initial assessment of the correlation between radionuclide concentrations and soil parameters was carried out using Spearman’s Rank. Those values with absolute $r_s > 0.8$ were considered suitable for use in linear regression prediction models. Explanatory variables to include in prediction models were derived using the Leaps package, with the best subset variables selected based upon the highest R^2 and lowest Mallow’s C_p values. Prediction models were developed by randomly splitting the whole dataset ($n = 90$) into training and validation datasets (Van Der Klooster et al. 2011) derived from 80 and 20% of the data, respectively. Model uncertainties were calculated using root mean square error (RMSE) and mean absolute error (MAE) for training and validation datasets (Heggemann et al. 2017; Pätzold et al. 2020). Training and validation datasets were checked to ensure that a representative range of values was used. The process of training and validation was repeated with different subsets of randomly selected data, and the range of model performance is reported here.

3 Results

Radionuclide activity concentrations (Bq/kg) and soil parameter summary statistics are shown in Table 1. Typical gamma counting uncertainties (relative standard deviation) were 13%, 30% and 20% for ⁴⁰K, ²³⁸U and ²³²Th, respectively. ¹³⁷Cs activity concentrations were lower than those of the geogenic radionuclides, and with lower energy emission, data were characterised by much greater counting uncertainties of around 50%. Measurements for ¹³⁷Cs were therefore not considered for further use in this study.

Table 1 Radionuclide activity concentrations and soil characteristics

	⁴⁰ K Bq/kg	²³⁸ U Bq/kg	²³² Th Bq/kg	¹³⁷ Cs Bq/kg	w %	LOI %	Sand %	Silt %	Clay %	TC %	TN %	BD g/cm ³
Mean	390	24	31	10	36	14	29	70	1.1	5.8	0.55	1.0
Min	209	15	21	4.0	17	7.0	17	45	0.52	2.5	0.21	0.70
Max	556	45	43	17	50	24	54	81	1.8	11	0.96	1.4

w moisture content, LOI loss on ignition, TC total carbon, TN total nitrogen, BD bulk density

Correlation between soil parameters and radionuclide concentrations showed negative associations between radionuclides and moisture, LOI, TC and TN, particularly with regard to ^{40}K ($r_s > -0.8$) (Fig. 2). Association between radionuclides and bulk density and texture, whilst significant, generally showed coefficients < 0.7 and were therefore not considered for use in prediction models in this study. Of those parameters showing strong association with one or more radionuclides, LOI, TC and TN most often require laboratory analysis of samples. Moisture, whilst obviously an important parameter, can be routinely assessed in situ. Loss on ignition is used as a proxy for organic matter (OM), and whilst this is useful in an agricultural setting, the derivation of meaningful concentrations of soil carbon requires the use of conversion factors, and, therefore, LOI was not used in prediction modelling here. However, the ability to obtain rapid and cost-effective in situ measurements of soil carbon and nitrogen is likely to be considered highly desirable with regard to soil health management. TC and TN were therefore the parameters taken forward for the development of prediction models.

Prediction models were developed for both TC and TN using multiple linear regression. Selection of explanatory variables, based upon the highest R^2 and lowest Mallows' C_p , meant that both models included ^{40}K and ^{238}U . Repeated

runs of TC training models using different random selections of data returned adjusted R^2 values of between 0.75 and 0.77 ($p < 0.001$) and RMSE of around 1% TC (18% relative uncertainty). Application of the training models to validation datasets (i.e. data not included in the training model) returned MAE of 1.08 to 1.13% TC (17 to 20% relative uncertainty). For TN, training models returned adjusted R^2 values between 0.77 and 0.79 ($p < 0.001$) and RMSE of 0.09% TN (16% relative uncertainty). MAE values from the validation exercises were around 0.1% TN (ranging from 15 to 18% relative uncertainty). It is noteworthy that, for both TC and TN, the inclusion of ^{238}U in the models resulted in only marginal improvements in R^2 , with ^{40}K as the most influential variable. As such, modelling with only ^{40}K as an explanatory variable (Fig. 3) resulted in comparable RMSE and MAE values.

4 Discussion

In this study, negative associations between moisture and organic matter-related parameters (LOI, TC and TN) and radionuclide activity concentrations are likely to relate to (i) the diluting effect of organic matter in relation to the gamma-emitting parent material and (ii) the attenuating

Fig. 2 Spearman's Rank correlation matrix with correlation coefficient (r_s) values shown. All values significant ($p < 0.05$) with the exception of those crossed through

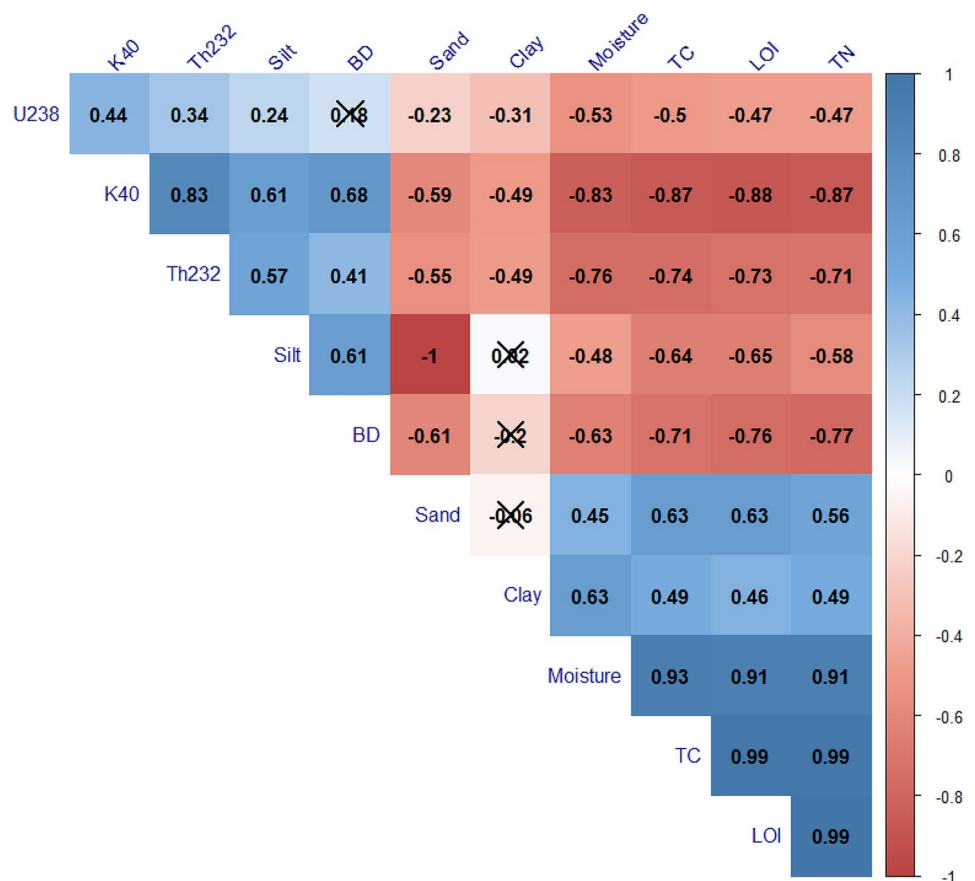
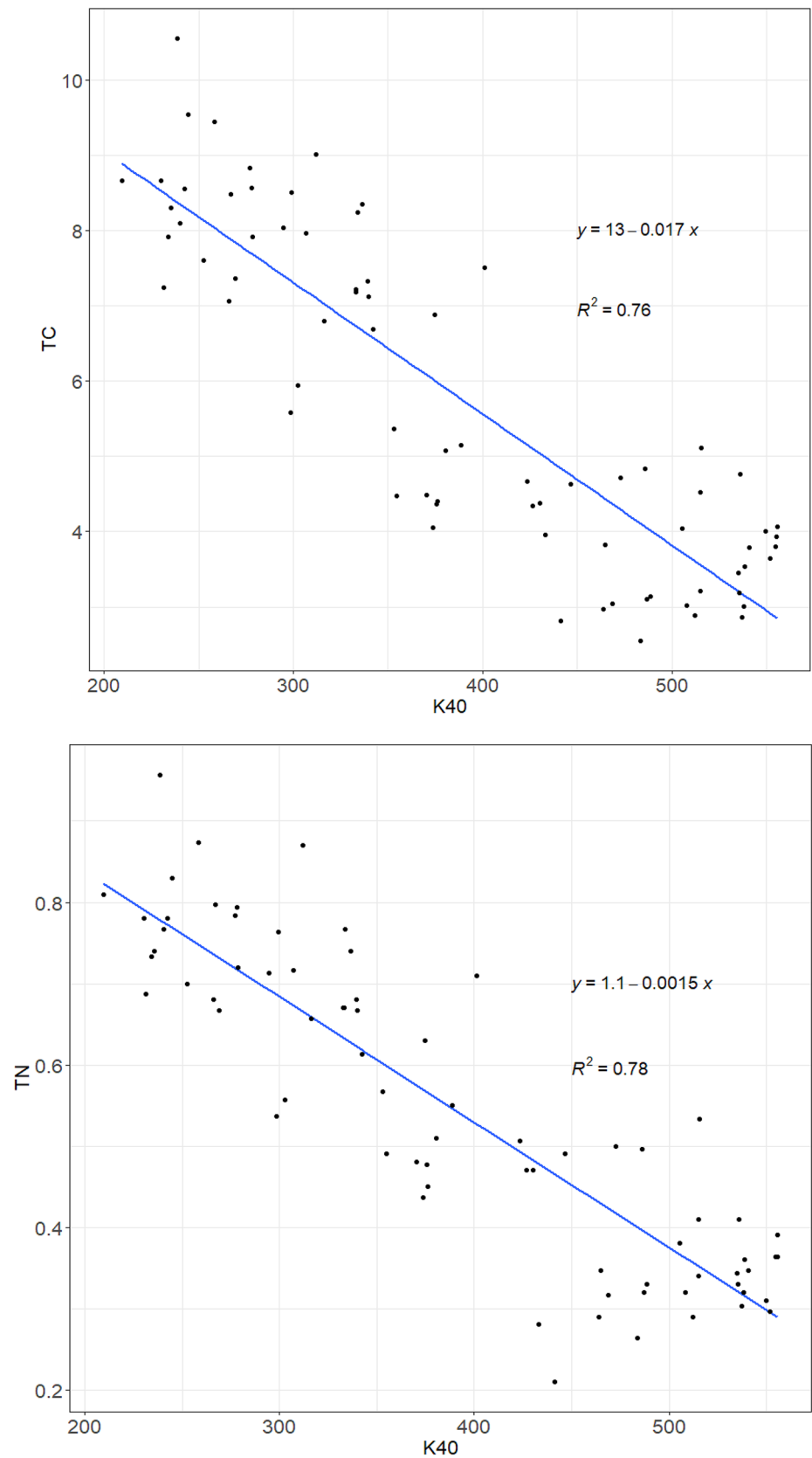


Fig. 3 Example training regression models for TC and TN (%) using ^{40}K (Bq/kg) as the explanatory variable



effect of soil moisture upon gamma rays, which is particularly pronounced for the lower energy ^{40}K emissions (Reinhardt and Herrmann 2019). The degree of data scatter (and RMSE) is likely to reflect an array of factors affecting detected gamma emission in situ. In this study, we chose to characterise the soil samples from the area directly below the detector, representing the area from which the greatest fraction of gamma radiation is detected (van der Veeke et al. 2021). In reality, however, gamma signals are likely to be detected from an unknown area around the detector, albeit at a lower intensity, adding to uncertainty in the sensor-to-laboratory comparison. For a detector height of 0.8 m above the ground, gamma signals are detected from a footprint with a radius of up to 14 m around the detector (van der Veeke et al. 2021). For the lower energy ^{40}K , this distance is likely to be less, potentially further influenced by changes in, for example, vegetation cover, soil moisture and mineralogy across this space. Data scatter is, therefore, unsurprising. However, the strength of the relationships shown with respect to ^{40}K is encouraging, especially in the context of informing decision-making based on concentrations of OM-related properties within a managed field unit.

To assess the use of these findings as a decision support tool, MAE should be considered in the context of soil management. With regard to TC for example, the maximum relative uncertainty from repeated validation (i.e. assessing actual versus predicted values with the validation datasets) was 20%. Considering this relative to a mean TC concentration of 5.8% (Table 1), the absolute error is 1.16% TC. As outlined above, inorganic carbon in the study soils was below detection limits, meaning that the TC values here represent total organic carbon. A generic target for building soil organic carbon stocks has been discussed in the context of the international *4 per thousand* initiative, launched during the 21st Conference of the Parties (COP 21) in Paris in 2015 (4 per 1000 Initiative 2020). This proposed a change in land management to support an increase of 0.4% soil organic carbon per year for 20 years. Poulton et al. (2018) explored this target and demonstrated that, in UK soil management, long-term studies show a wide range of annual SOC accumulation from < 0.4% up to 4.3%. Considering these values against the MAE shown here, detecting annual SOC increases with changing land use is plausible (where changes in SOC concentrations exceed uncertainties) but likely to be highly site-specific. However, in light of the range of TC values in this study, portable gamma spectrometry shows some promise in being able to rapidly detect areas of degradation or differences between land units where values in the upper and lower ranges are not within modelled uncertainties.

Considering the gamma sensor as a farm management support tool, rapid detection of SOC spatial patterns is valuable for identifying areas of degradation that can be targeted

for remediation approaches or change in use. From a regulatory and land management policy perspective, with growing interest in the role that soil management plays in carbon sequestration and its contribution to net zero ambition, parallel advances in complementary sensor technology, i.e. near infra-red spectroscopy (NIRS) (Shen et al. 2022), offer an opportunity for integrated sensor applications for soil health. In particular, we see huge potential in coupling high spatial resolution portable gamma sensor data with targeted in situ NIRS measurements to improve site-specific SOC estimates in support of carbon evaluation agendas. Advancement in autonomous agricultural robotic platforms (Duckett et al. 2018) offers further scope for rapid determination of soil health at a scale and resolution of maximum benefit to farm and landscape-level decision-making.

5 Conclusions

At a key time of agricultural transitioning and increasing attention upon soils for food, water and energy security, there is a need for rapid and cost-effective determination of soil health parameters. This study demonstrated the potential use of portable gamma spectrometry for monitoring soil carbon and nitrogen at a UK agricultural site. The study presents promising initial findings which can stimulate further study during this critical transition period, especially in the context of integrated sensor technology to support agricultural decision-making for improved soil health.

Further studies are required across a range of soil types and increasing spatial scales, using continuous monitoring approaches both on the ground and using unmanned aerial vehicles, to demonstrate this utility across a range of land management regimes and in different agroclimatic zones. The potential for developing generic models with more complex machine learning approaches that can be applied across different soil types and land units should be investigated.

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Author contribution All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Alex Taylor, Alice Kalnins, Martha Koot and Will Blake. The first draft of the manuscript was written by Alex Taylor, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability The datasets analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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